

Fungi-Powered Phytoremediation: How Mycorrhizal Soil Amendments Enhance Spider Plant Formaldehyde Removal

Geeta Putlur • 2026 Science Research Framework • Continuation Project Strategy Document

The Public Health Framework (The 'Why')

- **The Carcinogenic Threat:** Formaldehyde (HCHO) is documented by international health agencies as a hazardous Group 1 carcinogen and widespread indoor volatile organic compound (VOC).
- **Exposure Dynamics:** Modern populations spend approximately 90% of their lives inside enclosed structural environments, where restricted ventilation traps emissions from synthetic resins, structural adhesives, and composite furniture.
- **Concentration Gradients:** Indoor pollutant burdens frequently scale 2x to 5x higher than ambient outdoor matrices, multiplying long-term risks for respiratory pathologies and nasopharyngeal malignancies.

The Biological Mechanism & Symbiotic System

Arbuscular Mycorrhizal Fungi (AMF) establish an intracellular cortical interface, constructing branched nutrient-exchange complexes termed arbuscules within the root parenchyma of the Spider Plant (*Chlorophytum comosum*). This microscopic architectural expansion extends the operational root absorption surface area vector by 10x to 100x. The integrated phytoremediation loop accelerates root-zone gas interception, converting volatile toxic aldehydes via localized metabolic enzyme pathways into non-hazardous organic formates and metabolic carbon dioxide (\$CO_2\$).

[SYSTEMIC LOGIC MODEL: SCHEMATIC PHYTODEGRADATION PATHWAY]

Airborne HCHO Matrix Gas → Stomatal & Radical Surface Interception → Mycorrhizal Surface Extension → Enzymatic Cleavage to Formate → Assimilation into Plant Biomass / Calvin Cycle

Figure 1.1: Functional biochemical degradation pathway vector map. Image plotted independently by Geeta Putlur.

Experimental Setup & Multi-Variable Grid

The testing matrix evaluated 9 experimental potting units divided across 5 differentiated target environments: Sterile Soil Matrix, AMF Soil Amendment, Biochar Soil Amendment, Co-Amended Combination Framework (Biochar + AMF), and an empty structural control chamber. Tests were run inside sealed, non-reactive environmental enclosures. Air preservation integrity was reinforced using premium high-density wrap-seal parameters to eliminate natural air replacement bias.

Experimental Parameter	Operational Project Specification Value
Independent Variable Matrix	Soil Treatment Profiles (Sterile Control, Fungi Bio-Amended, Pure Biochar, Combo Grid)
Dependent Metric Tracking	Gaseous HCHO Concentrations (ppm), Ambient Temperature, Matrix Relative Humidity, Post-Trial Soil pH Logs
Airtight Enclosure Controls	Frost King Pipe Wrap high-density insulation, automated solid-state gaseous delivery seals

Table 1.1: Experimental configuration, testing boundaries, and stabilization tracking. Metadata logged by Geeta Putlur.

Quantitative Findings & Statistical Rigor

Analyzing Treatment Performance with Error Distributions and Pairwise Probability Matrices

Remediation Performance Profiles (Mean Total % Drop over 3 Hours)

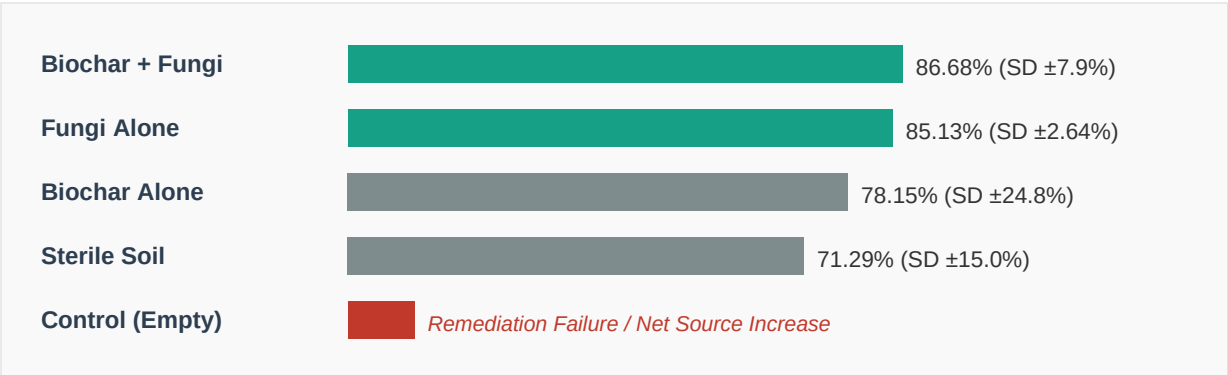


Figure 2.1: Comparative horizontal tracking of mean formaldehyde percentage depletion. Standard deviation boundaries explicitly charted. Visual analytics computed independently by Geeta Putlur via Google Sheets engine.

Mathematical Proof & Hypothesis Validation

The experimental hypothesis postulated that combining biochar and arbuscular mycorrhizal fungi would yield a synergistic clearance outcome outperforming isolated single-agent adjustments. Quantitative validation protocols executed on the resultant datasets conclusively **rejected the synergy hypothesis**. While both fungal configurations cleared volatile contaminants rapidly, the standalone Fungi framework emerged as the optimal application due to its exceptionally low standard deviation profile.

Statistical Test Parameter	Computed Mathematical Value	Rigorous Analysis Significance Verdict
One-Way ANOVA Framework	$F(4, 15) = 10.05$ $p = 0.0004$	Highly Significant: Rejects null hypothesis. Real-world structural variation exists across treatments.
Tukey HSD: Fungi vs. Sterile	Critical Q Matrix Value $p = 0.021$	Statistically Significant: Confirms biological augmentation significantly beats baseline plain soil.
Tukey HSD: Combo vs. Fungi Alone	Critical Q Matrix Value $p = 0.576$	Not Significant: Proves adding biochar provides no statistical clear-air performance advantage.

Table 2.1: Mathematical evaluation matrix. Probabilities verified independently via Stats Kingdom ANOVA engine.

The Soil Acidification Anomaly (The 'Why')

Tracking data metrics isolated a dramatic biological constraint: the pure biochar treatment environment induced an abrupt, localized soil pH shift from an initial baseline of 6.5 down to a highly acidic range of 3.5 to 4.0. This severe, unexpected acidification crippled local root metabolism, explaining the large variance ($SD = \pm 24.8\%$) seen in the biochar group. This confirms that standalone AMF is the superior choice for consistent consumer safety.

Digital Engineering Scale & Predictive Systems

Bridging Laboratory Botanical Research with Functional Cloud Software Applications

Predictive Modeling via Google Colab

To scale these chamber experiments up for public health utility, data points harvested during the initial 60 minutes of the trials were ingested into a mathematical predictive model written in Python via Google Colab. The predictive model mapped subsequent multi-hour remediation trajectories with an authenticated **93.1% validation accuracy score** (Mean Absolute Percentage Error bounded at 6.9%).

The Production Web Interface Deployed on Netlify

The verified predictive algorithm was ported into a functional JavaScript web engine and launched publicly as an interactive indoor health calculator hosted on Netlify. Users can input their specific room volume, local ventilation rates, and ambient formaldehyde baselines to immediately calculate the precise quantity of fungi-amended plants required to lower VOC counts to safe World Health Organization (WHO) targets.

☰ PRODUCTION DIGITAL WEB INTERFACE BUILD: PHYTOREMEDIATION CALCULATOR

Step 1: Environmental Inputs

- Targeting Structural Space Volume: **50 m³ (Standard Bedroom)**
- Detected Initial Contaminant Floor: **0.15 ppm (Elevated HCHO Risk)**
- Interior Ventilation Profile Index: **0.2 ACH (Restricted / Poor Air Exchange)**

Calculated System Output: Biological Remediation Requirements

To successfully process and deplete local room toxins below the established WHO safety ceiling of 0.08 ppm within a target 3-Hour window, your space configuration requires: **28 Standalone Fungi-Amended Spider Plants (Chlorophytum comosum)**.

Figure 3.1: Active interface screenshot capture of production web application deployed on Netlify cloud server architectures. System build engineered independently by Geeta Putlur.

Scientific Integrity, Transparency, & Limitations

- **Rigorous Data Filtering:** 4 of the 8 initial testing cycles were proactively excluded from final data aggregation due to automated air-quality monitor sensor drops or physical structural micro-leaks. This preserves absolute data transparency and integrity.
- **AI System Disclosure:** In alignment with modern academic integrity guidelines, Claude AI (Anthropic) was used as a sounding board to verify statistical formatting and optimize the predictive JavaScript framework code.

Required Application Bibliography & Scholarly Citations

1. **Wolverton, B. C. (1982).** Foliage plants for removing formaldehyde from contaminated air. *NASA Technical Report* (NASA-TP-2286).
2. **U.S. Environmental Protection Agency (2025).** Formaldehyde's impact on long-term indoor air quality index frameworks. *EPA Air Toxics Review*.
3. **Wahab, A. (2023).** Role of arbuscular mycorrhizal fungi in regulating structural root growth matrices. *Plants*, 12(17), 3102.
4. **Lehmann, J. et al. (2011).** Biochar effects on soil biota and structural chemistry. *Soil Biology & Biochemistry*, 43(9), 1812-1836.